Arab Academy for Science, Technology & Maritime Transport

College of Engineering and Technology

EXAMINATION PAPER

Course Title: Microprocessors Systems **Date**: 10th December 2020

Course Code: CC 421 **Time allowed**: **1:30** Hr.

Lecturer: Dr. Hesham El Zouka

Student's Name : ----------

Reg.# : ----------

Question #	**Marks**	
	Available	Actual
1		
2		
3		
Total	**15**	
Lecturer	**Name**:	
	Signature :	
	Date: ----------	

7th Week Exam

Answer All of the following questions

Question #1:

A. Identify the choice that best completes the statement or answers the questions.

(1) If DS = D321H, then the range of physical addresses for the Data segment is:

A. 00000H – 0D321H
B. D3210H – D321FH
C. D3210H – E320FH
D. 0D321H - 1D320H

(2) What is the largest signed integer that may be stored in 16 bits?

A. $2^{16} - 1$
B. $2^{15} - 1$
C. 2^{16}
D. 2^{15}

(3) The ———— allows the program to run in a single stepping mode.

A. Carry Flag
B. Direction Flag
C. Trap Flag
D. Parity Flag

(4) What will be the contents of register AL after the following program has been executed

```
MOV AL, 8C
MOV BL, 7E
ADD AL, BL
```

A. 0A and carry flag is set
B. 6A and carry flag is set
C. 0A and carry flag is reset
D. 6A and carry flag is reset
E. None of the above

(5) The MOV data transfer instructions will.

A. Affect flag register
B. Not affect flag register
C. The control flags
D. None of the above

(6) Which flag does the 8086 / 8088 use to check for unsigned arithmetic overflow?

A. OF
B. SF
C. CF
D. None of the above

(7) The stack pointer is associated with ———— By default

A. ES
B. DS
C. SS
D. CS

(8) Which register are used as an offset address for the source address string instruction in the microprocessor?

A. SI
B. DX
C. DI
D. BX

(9) The stack memory is represented by two registers

A. BP, DI B. SS,SP C. CS,SP D. None of the above

(10) A 24 - bit address bus allows the processor to access a memory space of :

A. 256 KB (b) 1 MB (c) 4 MB (d) 16 MB

B. Indicate whether each of the following statements is TRUE or FALSE and correct the false ones:-

(1) The Intel x86 CPU is known as a CISC processor. ✓

(2) The INT instruction can be hardware or software generated. ✓

(3) A PUSH increments SP by 2, and a POP decrements it. ✗

(4) The statement MOV AX, [SI + DI] is legal. ✗

(5) The following instruction will produce FFFCH in AX: ✓

MOV AX, -4

C. What will be the values of the specified registers and flags after the execution of the following instructions?

a)	b)

STC
MOV AX,8A03H
MOV CX,9C0AH
ADD AX, 1234H
ADC CX,AX 9C0A + 9C37
 	
AX= (9C37)$_{16}$, CX= (3841)$_{16}$
 	
CF=1, PF= ,AF= ,SF=0 ,ZF=0 ,OF=
	CLC
	MOV AX,2F36H
	MOV DX,1234H
	SBB AL, DL
	ADC DH,AL
	36 - 34
	12 + 02
	
	AX= (2F02)$_{16}$, DX= (1434)$_{16}$
	
	CF=0, PF= ,AF= ,SF=0 ,ZF=0 ,OF=0

3841

D. Use the following data for the next questions:

```
,data
    val1  DQ 1234ABCDH
    val2  DQ ?
```

(i) Consider EAX=07FE3C5DH, write an instruction that increment register AX by one.

(ii) Write an assembly program that adds the negative value of EAX to val1, and store the result in val2.

Good Luck.